

3. Memory transfers of output vectors from the device memory to the host memory.

Addresses of input and output vectors are passed as arguments to the CUDA kernel. In contrast, FASTCUDA runs everything on the same chip, thus favouring a different memory model where all the threads of a kernel and the host program can share a single global memory. In this model, hardware kernels inside the accelerator have direct access to the memory in order to read their input vectors and write output vectors.

Implementing CUDA kernels in hardware

In FASTCUDA, the code of hardware kernels is preprocessed before synthesis. To aid in this preprocessing the programmer is required to use '#pragma' directives in order to specify which

ones among the kernel arguments are inputs and outputs, as well as their sizes. The result of translation from CUDA to SystemC is given below:

```
//SystemC module\

SC_MODULE (addMod) {
    sc_in<int> A, B, C, threadIdx_x;
    sc_port<fcMem> sMem, gMem;
    sc_in<bool> clk, start;
    sc_out<bool> done;
    SC_CTOR (addMod) {
        SC_CTHREAD (add, clk);
        reset_signal_is (start);
    }
    // kernel
    void add() {
        int i = threadIdx_x;
        gMem.writeFloat (C+i,
            gMem.readFloat (A+i) +
                gMem.readFloat (B+i));
    }
}
```

SystemC interface called 'fcMem' is provided, like in latest GPUs,

to better exploit the AXI interface bandwidth.

Potential applications

Heterogeneous systems use advances in computing architectures to deliver improvements in power efficiency, performance, programmability and portability across computing devices. However, with heterogeneous architectures, care is required by the programmer to fully utilise hardware. Potential applications include smartphones, wearables, drones, high-end cameras, human-machine interfaces, industrial automation platforms, and security and surveillance products. The architecture has the potential to accelerate design of mobile and IoT edge devices, as it can easily handle features like voice, gesture and image recognition, haptics, graphics acceleration and signal aggregation. **EFY**

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